steady_drift_plus

equity strategy | family: equity-longonly | status: research | derived from `steady_drift`

What this strategy does

Steady Drift with smaller, more numerous positions, 80% invested cap and a 12% risk target (more return, same or better Sharpe)

Why it should work. Breadth lowers the 5%-of-volume cap's bite (56% of positions capped -> 39%) and diversifies, so the same signal supports a higher risk target without a lower Sharpe.

Design notes (from the strategy's source file).

Steady Drift -- a long-only, never-borrowing PEAD book with a constant-risk throttle (research).

What it does * Buys only the best earnings surprises (top decile of the point-in-time SUE rank) in stocks that trade at least $2M a day, holds each for exactly 60 trading sessions, and never sells short. * Sizes each position by how calm the stock is (inverse idiosyncratic variance, relative to the typical stock of the past year), at most 2% of NAV per name and 5% of the stock's daily dollar volume. Positions are admitted in announcement order, so no decision depends on events that have not happened yet. * Total stock cost basis is capped at 70% of NAV; the rest is cash earning the Treasury-bill rate (not the index). * A risk throttle scales the whole book down when its recent volatility exceeds the target (never above 100% invested, so it can only reduce risk, never borrow).

Why: after the methodology fixes, every market-neutral book collapses to 1-3% a year once short collateral is counted (they tie up 1.2-5.6x NAV), and the options edge disappears at real bid/ask. What survives is the long side of the surprise signal; docs/STEADY_DRIFT.md and docs/META_STRATEGY.md carry the evidence and the shuffled-surprise placebo.

STATUS research: the design was chosen after seeing 1996-2013; a shuffled-surprise test shows about half of the excess return over the cap-weighted index is the event universe (small/mid-cap, low-vol), not surprise skill; the T-bill series is typed from memory (pead/core/rates.py).

parameter	value
base_unit_fraction	0.003
max_gross_leverage	0.8
liquidity_cap_frac	0.05
return_basis	raw
cap_mode	chronological
exposure_cap	open_book
beta_overlay	0.0
hold_sessions	60
long_only_cash	True
risk_target_vol	0.12
name_cap_frac	0.01

Performance

metric	steady drift plus	market (CRSP VW)
Annualized return	12.07%	9.94%
Annualized volatility	9.97%	19.35%
Sharpe (NAV-pct)	1.19	0.59
Sortino	1.73	
Max drawdown	-22.1%	-55.5%
Calmar	0.55	
Final NAV ($10M start)	$91,947,400	
Worst day	-5.27%	
% positive days	59.9%	
Sharpe 1996-2001	1.71	
Sharpe 2002-2007	1.32	
Sharpe 2008-2013	0.79	

Correlation of daily returns with the market: **0.87**.

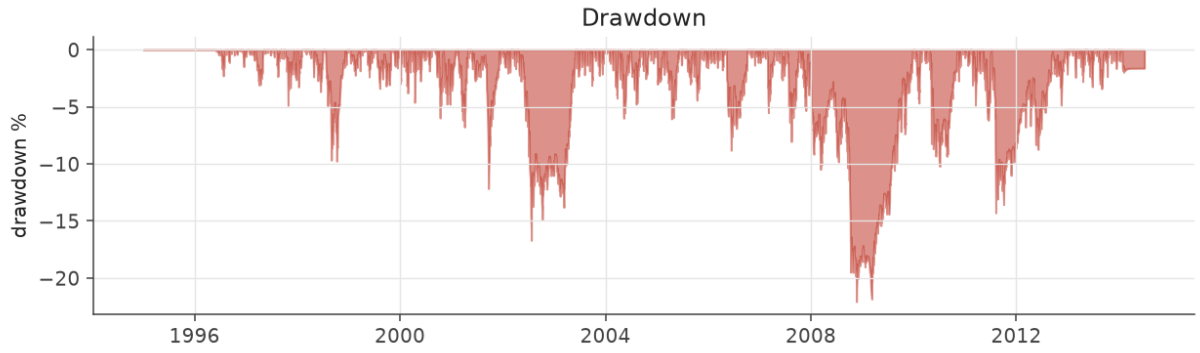
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.19

• best return among equity: this strategy; this one: 0.121

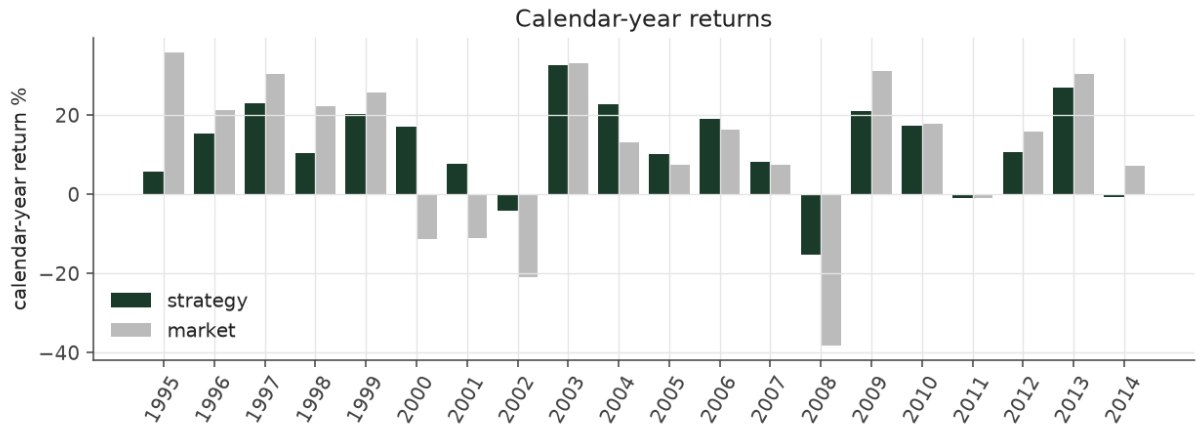
• best Calmar among equity: `steady_drift_calm` (0.601); this one: 0.547



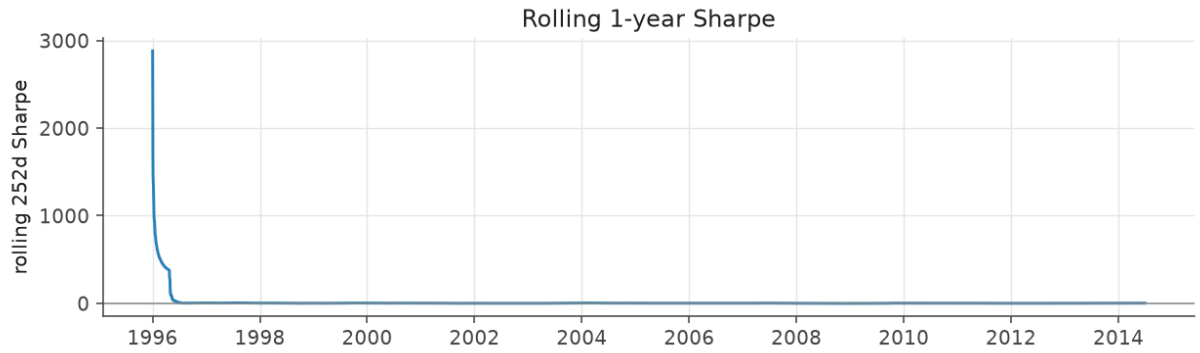
NAV (log scale)



Drawdown from running peak

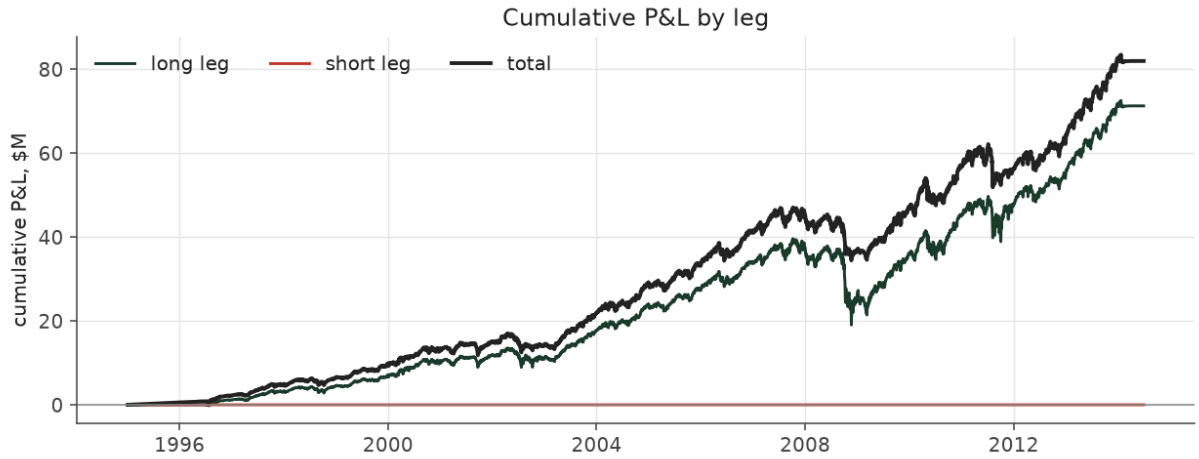


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

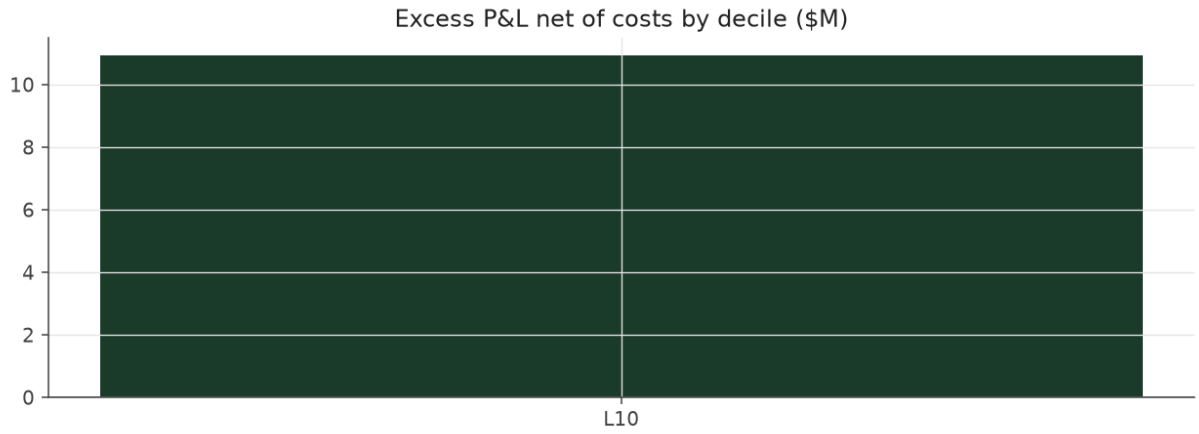
	long leg	short leg
P&L ($M, before costs)	71.25	0.00
P&L after costs ($M)	68.75	0.00
of which common (average-event) component ($M)	21.73	0.00
of which excess / PEAD component ($M)	13.44	0.00
excess minus costs ($M)	10.93	0.00
gross return per $ traded (bp)	409.80	-
positions in the money (hit rate)	0.64	-
stand-alone Sharpe (leg only)	0.89	-

Daily P&L correlation between the two legs: **nan**.

• The two legs' daily P&L are nan correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.89 long / nan short vs 1.19 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	11,961	1,738.64	68.75	13.44	2.50	10.93	100.0%	62.88



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	107	4.20	0.10	-0.10	0.01	-0.11	-1.0%	-263.64
2	1,241	79.04	4.36	1.59	0.19	1.40	12.8%	177.01
3	2,655	260.94	13.44	4.11	0.52	3.59	32.8%	137.58
4	3,838	565.80	22.18	4.79	0.87	3.92	35.9%	69.32
5	4,120	828.65	28.67	3.04	0.91	2.13	19.5%	25.73

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
2	813	85.47	3.17	0.56	0.26	0.30	2.7%	34.92
3	2,983	363.99	18.75	7.95	0.73	7.22	66.0%	198.37
4	3,916	573.11	21.45	2.54	0.80	1.74	15.9%	30.38
5	4,249	716.07	25.39	2.39	0.72	1.67	15.3%	23.37

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
41-60d	11,961	1,738.64	68.75	13.44	2.50	10.93	100.0%	62.88

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	3,486	192.78	11.46	5.28	0.27	5.01	45.8%	259.88
2002-07	4,120	606.29	21.94	8.98	0.85	8.13	74.4%	134.11
2008-13	4,355	939.57	35.34	-0.83	1.38	-2.21	-20.2%	-23.50

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	2,809	288.02	12.51	2.13	0.41	1.72	15.7%	59.54
Chems	288	62.63	3.17	1.73	0.09	1.64	15.0%	262.21
Durbl	273	39.95	0.94	-0.32	0.06	-0.37	-3.4%	-93.58
Enrgy	348	43.29	2.59	1.09	0.06	1.03	9.4%	237.33
Hlth	990	138.09	4.90	0.24	0.19	0.05	0.5%	3.66
Manuf	1,412	231.22	7.95	2.45	0.32	2.13	19.5%	92.10
Money	1,764	369.82	14.18	0.86	0.54	0.32	2.9%	8.57
NoDur	578	90.93	3.65	0.61	0.13	0.49	4.5%	53.66
Other	1,557	201.85	8.95	3.41	0.31	3.10	28.3%	153.40

Shops	1,209	146.91	4.85	0.58	0.21	0.36	3.3%	24.76
Telcm	304	30.39	1.16	-0.03	0.04	-0.07	-0.7%	-23.55
Utils	429	95.52	3.90	0.69	0.14	0.55	5.0%	57.48

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	4,426	263.62	8.14	0.51	0.36	0.15	1.4%	5.74
low vol	3,512	903.62	39.11	8.70	1.33	7.36	67.3%	81.49
mid vol	3,841	561.50	21.46	4.42	0.80	3.62	33.2%	64.56

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	12.07%	0.00%
0.5%	0.0	12.07%	0.00%
1.0%	0.0	12.07%	0.00%
2.0%	0.0	12.07%	0.00%
5.0%	0.0	12.07%	0.00%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.